Bt201 notes for final term

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1. To assemble a genome from a series of sequences requires tremendous **computational power.**
2. The first genome to be sequenced completely was that of **Haemophilus influenzae.**
3. The **laborious edman degradation** technique has now been replaced mostly by sequencing using mass spectrometry.
4. Species differ from foundation species in they are more likely to belong to **higher trophic levels**

5. Annelids range from **carnivores, herbivores, scavengers, deposit feeders, and filter feeders.**
6. Quinine – drug used to treat **malaria**
7. Modern bread wheat is a mixture of recombined genes from **three different wild species.**
8. One of the first questions that is asked when a species is considered for whole-genome sequencing is, **what is the size of its genome?**
9. **Allelic richness** Is the number of alleles in a sample (population),
10. On islands and habitat patches on continents**, species richness increases with area and decreases with isolation.**
11. Succession in the intertidal zone **requires 1 to 3 years**.
12. Highest rates of terrestrial primary production occur under warm(temperatures) , moist conditions .
13. **Heterogeneity**, in a community's environment, this may allow for greater species richness because there are more distinct habitats to be occupied.
14. The refuges that promote the persistence of hosts and prey include secure places to which the exploiter has **limited access**.
15. The environment limits population growth **by changing birth and death rates.**
16. An environment in which resources are patchy also fosters **clumped distributions**.
17. This more complex level of social behavior, which is considered to be the pinnacle of social evolution, is called **eusociality.**
18. The most familiar relative measure of the water content of air is **relative humidity**.
19. The rate at which organisms can take in energy is **limited,**
20. The earth's ecosystem interacting with the physical environment as a whole to maintain a steady state system intermediate in the flow of energy between the high energy input of the sun and the thermal sink of space is called **Biosphere.**
21. Multicelled algae appear as carbonaceous filaments from **1.8 billion-year**-old rocks in China.
22. Human is in **Primate** order:
23. Maximum parsimony assumes that **the tree that requires the fewest evolutionary events** is most likely.
24. In the 18th century, **Carolus Linnaeus** published a system of taxonomy based on resemblances
25. Oparin (1924): "**Primeval soup**" theory
26. **Haldane (1929):** "Hot, dilute soup" theory.

27. Horizontal Gene Transfer may be **Conjugation, Tranformation, and Transduction.**
28. Which is not the Positive interaction? **Competition**
29. **Disruptive Selection**: natural selection that favors two or more extreme phenotypes are favored over any intermediate phenotype
30. The advantage of in situ conservation is not one of following: **The species will not evolve in their environment.**
31. The most common extensions of genomics are EXCEPT? **(Ans) vaccinomics.**
32. Vertebrates make only--------of all animal species? **(ans) 5%.**
33. One of the following is not how protozoa reproduce? **Parthenogenesis**.
34. Gymnosperm means? Ans **naked seeds.**
35. Supporting evidence for end symbiosis comes from …… **by studying prokaryotic cell** .
36. **Meanwhile, Smith, Buckland** and Cuvier himself (along with Brongniart) were developing paleontology.
37. Population include variation in… genetic and phenotypic .
38. **Clausen, Keck, and Hiesey**…. explored the extent and sources of morphological variation in plant populations, including both the influences of environment and genetics.
39. Hardi weinberg condition are **…. Random mating, no mutations, large population size, no immigration, all are right answers**.
40. **Frankham** showed that compared to mainland populations, island populations generally include less genetic variation.
41. **Ecologists**-- now have many powerful tools, ranging from classical techniques to modem technologically sophisticated approaches, for assessing the extent of genetic variation within and among populations and meeting the challenge of documenting and conserving biodiversity.
42. An environment in which resources are patchy also fosters **clumped distributions**.
43. The environment limits population growth by **changing birth and death rates.**
44. The intrinsic rate of increase, rm is **the maximum rate** of increase for a given species.
45. **Lotka voltera** independently expanded the logistic model of population growth to represent interspecific competition
46. Field experiment feature is **knowledge of initial conditions**.
47. The refuges that promote the persistence of **hosts and prey** include secure places to which the exploiter has limited access.

48. **Frank Preston** (1948) graphed the abundance of species in collections as distributions of species abundance, with each abundance interval twice the preceding one
49. A **food web** summarizes the feeding relations in a community.
50. **Robert Paine** (1966) proposed that the feeding activities of a few species
51. **Primary succession** occurs on newly exposed geological substrates not significantly modified by organisms.
52. Succession in the intertidal zone requires **1 to 3** years
53. **Repeat photography** can be used to detect long-term ecological change
54. The local population density of the Glanville fritillary **butterfly.**
55. In the Sonoran Desert, plant distributions map clearly onto soils of different ages and form a **vegetative** mosaic that closely matches soil mosaics
56. On islands and habitat patches on continents, **species richness** increases, with area and decreases with **isolation**.
57. In the 18th century, Carolus Linnaeus published a system of taxonomy based on resemblances
58. Genetic diversity means that the population contains most of the possible _____for each particular gene locus.
    a) Genes
    b) Alleles
    c) Allelic frequency
    d) Genetic frequency
59. Losses of genetic diversity can imply_____.
    a) lack of evolution
    b) premature extinction
    c) both a and b
    d) none
60. Fitness _____with reduced genetic variation.
    a) Decreases
    b) Increases
    c) Remain the same
    d) Both a and b
61. Populations of endangered species are small and tend to _____genetic diversity.
    a) Gain
    b) Lose

c) Share
d) None

62. Physical location of a gene is called_____.
a) Allele
b) Gene point
c) Locus
d) None of the above

63. Differentiate between the Homozygous and Heterozygous?
Homozygous:
Two of the same alleles at a given locus.
Heterozygous:
Two different alleles at a given locus.

64. Contribution of an individual's genotype to the next generation is called _____.
a) Copy
b) Transmission
c) Fitness
d) None

65. Define Gene pool?
Different individuals carry different genetic information; The sum of the variation of an interbreeding population is called Gene pool.

66. Census population size (Nc) vs. Effective population size (Ne), Usually Ne is much smaller that Nc _____.
a) 10 to 20%
b) 10 to 30%
c) 20 to 30%
d) 20 to 40%

67. Enlist the Four factors responsible for genetic diversity loss in small populations?
The four factors rev are:
- Genetic drift,
- Founder effect,
- Demographic bottleneck,
- Inbreeding

68. What is Founder effect define it?

"The establishment of a new population by a few original founders which carry only a small fraction of the total genetic variation of the parental population." (Ernst Mayr), Alleles may be lost, Differentiation from the parental population, Example: Irish Potato Famine.

69. Write the Usefulness of Ecosystem Management and Genetics in Conservation?

Ecosystem Management and Genetics in Conservation Useful in

a. Endangered small populations,
b. Captive breeding,
c. Translocation of individuals,
d. Determining dispersal patterns,
e. Recognize limitations and scale of application,
   - It's easy to lose the sight of the larger picture and try to apply a genetics approach when it is not appropriate to do so.
   - Ecosystem Management and Genetics in Conservation is Very powerful tool
   - Usually useful to solve specific problems
   - Rarely useful if it is not combined with other type of approaches.

70. what is intermediate hypothesis suggests?

- The intermediate disturbance hypothesis suggests that communities with a medium (intermediate) level of disturbance may have greater species diversity than communities with very frequent or very rare disturbances.

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73. **What is optimal theory?**

Optimal foraging theory:

Optimal foraging theory attempts to model how organisms feed as an optimizing process. Evolutionary ecologists predict that if organisms have limited access to energy, natural selection is likely to favor individuals that are more effective at acquiring energy and nutrients. Many animals select food in a way that appears to maximize the rate at which they capture energy.

74. Write four resons behind the fossils history uncomplete?

The fossil record is relatively incomplete for several reasons:

1. Soft tissues are rarely preserved
2. Movement of the earth's crust has obliterated and/or covered many fossils
3. Fossilization takes place only in certain types of habitats and favorable environments
4. Paleontologists have not dug up every place on earth

75. Specie richness and diversity richness?

Species richness is the number of different species in a particular community. If we found 30 species in one community, and 300 species in another, the second community would have much higher species richness than the first.

Species diversity is a measure of community complexity. It is a function of both the number of different species in the community (species richness) and their relative abundances (species evenness). Larger numbers of species and more even abundances of species lead to higher species diversity.

76. How the exicot predetrs distrub the food web?

Exotic predators can collapse and simplify the structure of food webs. Introduced fishes have overcome the native fishes of Lake Atitlan and Gatun Lake in Central America.

Examples:

The influence of the Nile perch on the fish community of Lake Victoria is enmeshed with massive changes in the lake's ecosystem. Introduction of the

Nile perch is rapidly reducing the species-rich fish fauna of Lake Victoria to a community dominated by a handful of species.

77. Evoloitnry history of human?

HUMAN EVOLUTION

Humans are too complex to be "understood" by any one field. Thus we will look at a few major steps in evolution and some of the things affecting human evolution. Humans are members of the order Primates which consists of about 180 species (there are 17 different orders of mammals which diverged 80-65 million years ago). Primates are a relatively old order of mammals. Most of the synapomorphies of this order are associated with an arboreal way of life: flexible digits, forward facing eyes, vision as a primary sense. These traits may have played a role in the evolution of brain size in the lineage leading to humans. Humans are a member of the family Hominidae which is believed to have diverged about 5 million years before the present (mybp) from the other members of the Old world monkeys. At least 20 mybp the Hominoids split off from the other old world monkeys. The dates are rough and get changed now and then.

78. What is hybrd weinberg principle?

"In a population mating at random in the absence of evolutionary forces, allele frequencies will remain constant."

79. How mustulim took place and write human mulusim wth other species.

80. Mutualism, interactions between individuals that benefit both partners, is a common phenomenon in nature that has apparently made important contributions to the evolutionary history of life and continues to make key contributions to the ecological integrity of the biosphere.

Humans in Mutualism:

Humans have developed a variety of mutualistic relationships with other species, but one of the most spectacular is that between the greater honeyguide and the traditional honey gatherers of Africa. In this apparently ancient mutualism, humans and honeyguides engage in elaborate communication and cooperation with clear benefit to both partners.The mutualism offers the human side a higher rate of discovery of bees' nests, while the honeyguide gains access

to nests that it could not raid without human help. Careful observations have documented that the honeyguide informs the honey gatherers of the direction and distance to bees' nests as well as of their arrival at the nest.

81. Why Lallys theory failed explain?

Lyell's mistake:

As a part of his general resistance to direction/ substantial change in geology, Lyell maintained there was no real direction of change in the fossil record either. This was greeted with pretty general astonishment, but Lyell claimed the apparent changes were just due to our having bad (misleading) samples...more mammals would soon appear in older formations, etc., as we extend our fossil collections. Within ten years (1840) it was clear that this was completely untenable.

82. Write phyla of ameboid protozon and write mode of locomotion?

The Protozoans (Ciliates Amoeboid Protozoans Flagellated Protozoans)

Kingdom Protozoa

Defining Characteristics

All are unicellular eukaryotes locomotion

- Amoeboid movement
- Swimming movement
- Gliding movement
- Metabolic movement

83. Write significance of genetic life tree?

Investigating the Tree of Life

Phylogeny is the evolutionary history of a species or group of related species. The discipline of systematics classifies organisms and determines their evolutionary relationships. Systematists use fossil, molecular, and genetic data to infer evolutionary relationships Phylogenies show evolutionary relationships. Taxonomy is the ordered division and naming of organisms

84. What are the processes of the ecosystem influenced by landscape structure

Landscape structure influences processes such as the flow of energy, materials, and species between the ecosystems within a landscape.

85. How nutrient enrichment says the human is altering the aquatic and terrestrial ecosystem?

Nutrient enrichment by humans is altering aquatic and terrestrial ecosystems. Nitrate concentration and export by the earth's major rivers correlate directly with human population density. Human disturbance also increases export of phosphorus from aquatic catchments.

86. How organism evolutionary history is documented in its genome.

An organism's evolutionary history is documented in its genome. Comparing nucleic acids or other molecules to infer relatedness is a valuable tool for tracing organisms' evolutionary history. DNA that codes for rRNA changes relatively slowly and is useful for investigating branching points hundreds of millions of years ago. mtDNA evolves rapidly and can be used to explore recent evolutionary events

Gene Duplications and Gene Families

Gene duplication increases the number of genes in the genome, providing more opportunities for evolutionary changes. Like homologous genes, duplicated genes can be traced to a common ancestor. Orthologous genes are found in a single copy in the genome and are homologous between species. They can diverge only after speciation occurs. Paralogous genes result from gene duplication, so are found in more than one copy inthe genome

87. Define ecology in our definition and write work of five different scientists.

What is ecological genomics?

"A scientific discipline that studies the structure and functioning of a genome with theaim of understanding the relationship between the organism and its biotic and abiotic environments." proposed the word genome: 1920 the German botanist Hans Winkler. The term genomics : mid-1980s and was introduced in 1987 as the name of a new journal (McKusick and Ruddle 1987).The suffix -ome :the collectivity of units (Lederberg and McCray 2001), example in coelome, the system of body cavities, The sequencing of rice was motivated by the huge importance of this species as a staple food for the world population (adam 2000). Zhou et al. (2004) proposed a classification of genomics

88. community structure

Community structure is essentially the composition of a community, including the number of species in that community and their relative numbers^11startm superscript, 1, end superscript. It can also be interpreted more broadly, to include all of the patterns of interaction between these different Species.

Factore

1. The climate patterns of the community's location.
2. The geography of the community's location.
3. The heterogeneity (patchiness) of the environment
4. The frequency of disturbances, or disruptive events.
5. Interactions between organisms.

89. Whole genome sequencing significience?

Many bacteria and parasitic protists were chosen because of their pathogenicity to humans. Other bacteria and fungi were taken as genomic models because of their potential to cause plant diseases (phytopathogenicity). The sequencing of rice was motivated by the huge importance of this species as a staple food for the world population (adam 2000).

90. Lyell's Mistake on natural selection?

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• **Kerner** showed that the differences in growth form observed by him and Bonnier were the result of temporary adjustments to environmental variation and not hereditary changes in the experimental plants.

• **Turesson's**studies provided evidence for genetic differences among populations.

• **Clausen, Keck, and Hiesey** explored the extent and sources of morphological variation in plant populations, including both the influences of environment and genetics.

• **Case** determined that the best predictor of chuckwalla, Sauromalus, body length was average winter rainfall.

• **Tracy's** laboratory growth experiments indicated that variation in body size among chuckwalla populations is at least partly determined by genetic differences among populations.

91. Darwin's Theory of Evolution - Natural Selection

The branch of biology that deals with the relations of organisms to one another and to their physical surroundings the political movement concerned with protection of the environment.

Ecology:

Phylogenetic trees do show patterns of descent. Phylogenetic trees do not indicate when species evolved or how much genetic change occurred in a lineage. It shouldn't be assumed that a taxon evolved from the taxon next to it. Applications: whale meat sold illegaly in Japan.

92. What is intermediate hypothesis suggests?

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biotechnologists